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Appendix C.1. rank-one determinant route: Determinant Algebra with the $\frac{2}{3}$ Normalization

Consider the rank-one determinant route Lagrangian

$$\mathcal{L}_{\mathrm{ROD}} = \sqrt{-g}\Big(\sqrt{\det\big[\mathbf{1} + \tfrac{2}{3}\lambda\,\tau\,\mathcal{T}\big]} - 1\Big), \qquad \lambda \in \mathbb{R}. \tag{67}$$

Using $\sqrt{\det(\mathbf{1}+X)} = 1 + \frac{1}{2}\mathrm{Tr}X + \frac{1}{8}[(\mathrm{Tr}X)^2 - 2\mathrm{Tr}X^2] + \mathcal{O}(X^3)$ and $\mathrm{Tr}\mathcal{T} = 0$, the quadratic piece is

$$\begin{aligned}\delta^2\mathcal{L}_{\mathrm{ROD}} &= \sqrt{-g}\,\frac{1}{8}\Big[-2\,\mathrm{Tr}\Big((\tfrac{2}{3}\lambda\,\tau)^2\mathcal{T}^2\Big)\Big] = -\frac{1}{4}(\tfrac{2}{3})^2\lambda^2\,\sqrt{-g}\,\tau^2\,\mathrm{Tr}(\mathcal{T}^2)\\ &= -\frac{1}{4}\cdot\frac{4}{9}\cdot\frac{3}{4}\,\lambda^2\,\sqrt{-g}\,\tau^2 = -\frac{1}{12}\,\lambda^2\,\sqrt{-g}\,\tau^2.\end{aligned} \tag{68}$$

With $\tau^2 = 9\eta^2|\Sigma_\epsilon|$ and $\mathrm{sgn}(\Sigma_\epsilon)\,I_T = -6\eta^2|\Sigma_\epsilon|$,

$$-\frac{1}{12}\lambda^2\tau^2 = -\frac{1}{12}\lambda^2\,(9\eta^2|\Sigma_\epsilon|) = -\frac{3}{4}\lambda^2\eta^2|\Sigma_\epsilon| = \Big(\frac{\lambda^2}{8}\Big)\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T,$$

so that

$$\boxed{\delta^2\mathcal{L}_{\mathrm{ROD}} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\mathrm{ROD}}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{69}$$

At the bulk-density level one may take $J^\mu_{\mathrm{ROD}} \equiv 0$. For unified boundary diagnostics we adopt a common canonical representative J^μ_{can} for *all* three routes (Sec. C.4).

Appendix C.2. Closed–Metric Route: Rank–One Deformation Equals rank-one determinant route to $\mathcal{O}(\mathcal{T}^2)$

Take the rank–one deformation $\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{2}{3}\lambda\,\tau\,\hat{n}_\mu\hat{n}_\nu$ so that $\sqrt{-\tilde{g}} = \sqrt{-g}\sqrt{\det[\mathbf{1} + \frac{2}{3}\lambda\,\tau\,\mathcal{T}]}$ and define $\mathcal{L}_{\mathrm{CM}} \equiv \sqrt{-\tilde{g}} - \sqrt{-g}$. The same algebra gives

$$\boxed{\delta^2\mathcal{L}_{\mathrm{CM}} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\mathrm{CM}}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{70}$$

Thus rank-one determinant route and CM have the *same bulk* coefficient $A_\star$ and differ only by improvements.

Appendix C.3. PT–Even CS/Nieh–Yan Shadow: Quadratic Reduction (σ_ϵ)

Using $T^A \wedge T_A = d(e^A \wedge T_A) - e^A \wedge e^B \wedge R_{AB}$, applying $*$ and the scalar PT projector (A2), and evaluating after C1, the PT–even piece reduces at quadratic order to

$$\boxed{\delta^2\mathcal{L}_{\mathrm{CS}^+} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\mathrm{CS}^+}, \qquad A_\star = \frac{\lambda^2}{8},} \tag{71}$$

where the Nieh–Yan assignment (A5) reshuffles only boundary conventions inside the PT–even sector.

Appendix C.4. A universal Canonical Improvement at Quadratic Order

For route–by–route flux comparisons it is convenient to select the *same* improvement representative for all routes:

$$\boxed{J^\mu_{\mathrm{can}}[h^{\mathrm{TT}}] \;\equiv\; \frac{A_\star}{2}\,a^3\Big(h^{\mathrm{TT}}_{ij}\,\nabla^\mu h^{\mathrm{TT}}_{ij} - \delta^\mu{}_0\,h^{\mathrm{TT}}_{ij}\,\dot{h}^{\mathrm{TT}}_{ij}\Big),} \tag{72}$$

so that we *adopt the convention*

$$\delta^2\mathcal{L}_X = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\mathrm{can}}, \qquad X \in \{rank-onedeterminantroute, \mathrm{CM}, \mathrm{CS}^+\}. \tag{73}$$

Different representatives differ by $\nabla_\nu X^{[\mu\nu]}$ and yield identical integrated fluxes under A4.

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